# FUVEST 4

## FUVEST 2017 1ª FASE

# FOTO PARA QUESTÕES DE 19 A 21

**THE NEW YORK TIMES**, JUNE 30, 2016. ADAPTADO



# TEXTO PARA AS QUESTÕES DE 19 A 21

PLANTS NOT ONLY REMEMBER WHEN YOU TOUCH THEM, BUT THEY CAN ALSO MAKE RISKY DECISIONS THAT ARE AS SOPHISTICATED AS THOSE MADE BY HUMANS, ALL WITHOUT BRAINS OR COMPLEX NERVOUS SYSTEMS.

RESEARCHERS SHOWED THAT WHEN FACED WITH THE CHOICE BETWEEN A POT CONTAINING CONSTANT LEVELS OF NUTRIENTS OR ONE WITH UNPREDICTABLE LEVELS, A PLANT WILL PICK THE MYSTERY POT WHEN CONDITIONS ARE SUFFICIENTLY POOR.

IN A SET OF EXPERIMENTS, DR. SHEMESH, FROM TEL_HAI COLLEGE IN ISRAEL, AND ALEX KACELNIK, FROM OXFORD UNIVERSITY, GREW PEA PLANTS AND SPLIT THEIR ROOTS BETWEEN TWO POTS. BOTH POTS HAD THE SAME AMOUNT OF NUTRIENTS ON AVERAGE, BUT IN ONE, THE LEVELS WERE CONSTANT; IN THE OTHER, THEY VARIED OVER TIME. THEN THE RESEARCHERS SWITCHED THE CONDITIONS SO THAT THE AVERAGE NUTRIENTS IN BOTH POTS WOULD BE EQUALLY HIGH OR LOW, AND ASKED: WHICH POT WOULD A PLANT PREFER? WHEN NUTRIENT LEVELS WERE LOW, THE PLANTS LAID MORE ROOTS IN THE UNPREDICTABLE POT. BUT WHEN NUTRIENTS WERE ABUNDANT, THEY CHOSE THE ONE THAT ALWAYS HAD THE SAME AMOUNT.

# QUESTÃO 19

SEGUNDO UMA DAS CONCLUSÕES DOS EXPERIMENTOS RELATADOS NO TEXTO, AS PLANTAS DE ERVILHA DEMONSTRARAM

A) SENSIBILIDADE AOS GESTOS HUMANOS AGRESSIVOS.

B) TER SISTEMAS NERVOSOS COMPLEXOS.

C) GRAUS DISTINTOS DE TOLERÂNCIA À UMIDADE DO SOLO.

D) CAPACIDADE DE ESCOLHAS ADAPTATIVAS CONFORME O MEIO.

E) COMPORTAMENTO PREVISÍVEL NO PROCESSO DE FLORESCIMENTO.

# QUESTÃO 20

CONFORME O TEXTO, UM DOS ELEMENTOS DA METODOLOGIA EMPREGADA NOS EXPERIMENTOS FOI

A) O NÚMERO DE MUDAS PLANTADAS.

B) A TÉCNICA DE DIVISÃO DE RAÍZES.

C) A LOCALIZAÇÃO DOS VASOS NA ESTUFA.

D) A ESCOLHA DA VARIEDADE DE ERVILHA.

E) O ESPAÇAMENTO DAS SEMENTES NOS VASOS.

# QUESTÃO 21

DE ACORDO COM OS EXPERIMENTOS RELATADOS NO TEXTO, EM CONDIÇÕES ADVERSAS, AS PLANTAS DE ERVILHA PRIORIZARAM O CRESCIMENTO DE RAÍZES NOS VASOS QUE APRESENTARAM NÍVEIS DE NUTRIENTES

A) ABUNDANTES.

B) ESTÁVEIS.

C) BÁSICOS.

D) IDEAIS.

E) VARIÁVEIS.

# TEXTO PARA AS QUESTÕES 22 E 23

A STUDY CARRIED OUT BY LAUREN SHERMAN OF THE UNIVERSITY OF CALIFORNIA AND HER COLLEAGUES INVESTIGATED HOW USE OF THE "LIKE" BUTTON IN SOCIAL MEDIA AFFECTS THE BRAINS OF TEENAGERS LYING IN BODY SCANNERS.

THIRTY_TWO TEENS WHO HAD INSTAGRAM ACCOUNTS WERE ASKED TO LIE DOWN IN A FUNCTIONAL MAGNETIC RESONANCE IMAGING (FMRI) SCANNER. THIS LET DR. SHERMAN MONITOR THEIR BRAIN ACTIVITY WHILE THEY WERE PERUSING BOTH THEIR OWN INSTAGRAM PHOTOS AND PHOTOS THAT THEY WERE TOLD HAD BEEN ADDED BY OTHER TEENAGERS IN THE EXPERIMENT. IN REALITY, DR. SHERMAN HAD COLLECTED ALL THE OTHER PHOTOS, WHICH INCLUDED NEUTRAL IMAGES OF FOOD AND FRIENDS AS WELL AS MANY DEPICTING RISKY BEHAVIOURS LIKE DRINKING, SMOKING AND DRUG USE, FROM OTHER PEOPLES' INSTAGRAM ACCOUNTS. THE RESEARCHERS TOLD PARTICIPANTS THEY WERE VIEWING PHOTOGRAPHS THAT 50 OTHER TEENAGERS HAD ALREADY SEEN AND ENDORSED WITH A "LIKE" IN THE LABORATORY.

THE PARTICIPANTS WERE MORE LIKELY THEMSELVES TO "LIKE" PHOTOS ALREADY DEPICTED AS HAVING BEEN "LIKED" A LOT THAN THEY WERE PHOTOS DEPICTED WITH FEWER PREVIOUS "LIKES". WHEN SHE LOOKED AT THE FMRI RESULTS, DR. SHERMAN FOUND THAT ACTIVITY IN THE NUCLEUS ACCUMBENS, A HUB OF REWARD CIRCUITRY IN THE BRAIN, INCREASED WITH THE NUMBER OF "LIKES" THAT A PHOTO HAD.

**THE ECONOMIST**, JUNE 13, 2016. ADAPTADO

# QUESTÃO 22

SEGUNDO O TEXTO, COMO RESULTADO PARCIAL DA PESQUISA, OBSERVOU-SE QUE

A) FOTOS COM IMAGENS NEUTRAS PROVOCARAM MENOR IMPACTO DO QUE AS QUE RETRATAVAM COMPORTAMENTO PERIGOSO.

B) OS PARTICIPANTES MOSTRARAM TENDÊNCIA A "CURTIR" UMA IMAGEM QUE JÁ HAVIA RECEBIDO NÚMERO CONSIDERÁVEL DE "CURTIDAS".

C) OS ADOLESCENTES DEMONSTRARAM CERTO DESCONFORTO, QUANDO SOLICITADOS A AVALIAR FOTOS PRODUZIDAS POR ELES PRÓPRIOS.

D) AS TAREFAS PROPOSTAS AOS PARTICIPANTES APRESENTARAM LIMITAÇÕES, POR TEREM FOCO EXCLUSIVO NA REDE INSTAGRAM.

E) A METODOLOGIA ADOTADA NO EXPERIMENTO CONFIRMOU CONCLUSÕES DE ESTUDOS ANTERIORES SOBRE REDES SOCIAIS.

# QUESTÃO 23

CONFORME O TEXTO, A REGIÃO DO CÉREBRO QUE SE MOSTROU MAIS ATIVA, QUANDO DA ANÁLISE DOS RESULTADOS DA RESSONÂNCIA, CORRESPONDE A UM SISTEMA DE

A) MEMÓRIA RECENTE.

B) DEFESA.

C) RECOMPENSA.

D) REPETIÇÃO.

E) INIBIÇÃO.